

ANSI X12 850 - Purchase Order

What it is

The X12 850 is the standard EDI transaction set used to transmit a purchase order from a buyer to a seller: what's being ordered, how much, at what price, and where it should ship.

Note: This is an original educational summary of the transaction set's structure, written from publicly known segment names/purposes. It is not a reproduction of the official ASC X12 standard, which is licensed and sold by X12.org. For a certified implementation guide, purchase the official standard from X12 or your trading partner's EDI documentation.

Envelope structure (shared by all X12 transaction sets)

Interchange - ISA: Interchange Control Header - sender/receiver IDs, delimiters
Functional Group - GS: Functional Group Header - groups transactions of the same type
Transaction Set - ST: Transaction Set Header - marks the start of one 850 document
... - ...: (document-specific segments, see below)
Transaction Set - SE: Transaction Set Trailer - segment count, control number
Functional Group - GE: Functional Group Trailer
Interchange - IEA: Interchange Control Trailer

850-specific segments

BEG - Beginning Segment for Purchase Order: PO number, PO type, issue date
REF - Reference Identification: Additional reference numbers (dept, contract, etc.)
PER - Administrative Communications Contact: Buyer contact info
FOB - F.O.B. Related Instructions: Shipping terms
DTM - Date/Time Reference: Requested ship / delivery dates
N1 - Name: Party identification (Buyer, Vendor, Ship-To), often looped
PO1 - Baseline Item Data: One line item: SKU, quantity, unit price, UOM
PID - Product/Item Description: Free-text description of the line item
CTT - Transaction Totals: Count of line items in the transaction

Mapping to a modern REST equivalent

This structure maps naturally onto a JSON PurchaseOrder payload: BEG -> order header fields, N1 loop -> a businessEntity array (buyer/vendor/ship-to), PO1/PID loop -> an itemDetail array, CTT -> a derived count rather than a transmitted field.